

Problem sheet–8

Sampling

1. Comparison of decimation and low frequency sampling

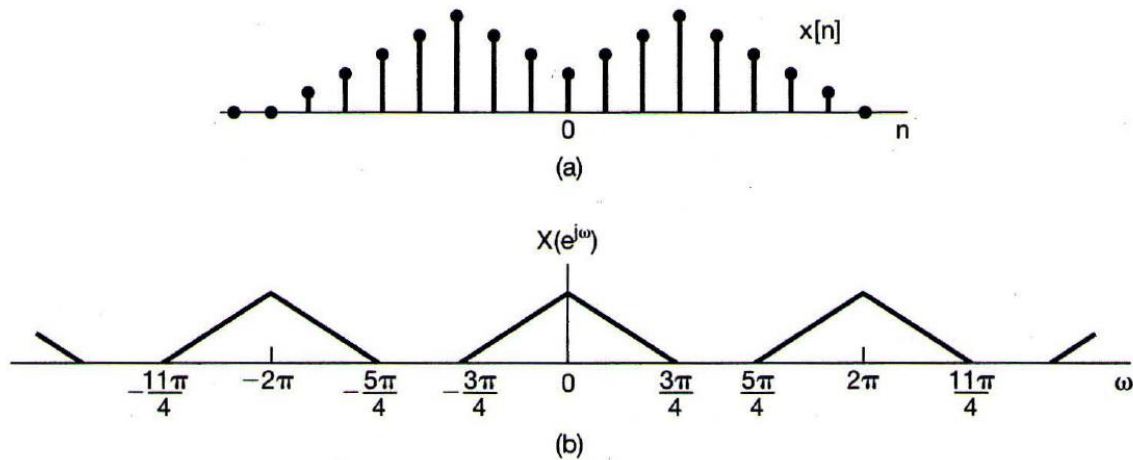
Consider a discrete-time sequence $x[n]$ from which we form two new sequences, $x_p[n]$ and $x_d[n]$, where $x_p[n]$ corresponds to sampling $x[n]$ with a sampling period of 2 and $x_d[n]$ corresponds to decimating $x[n]$ by a factor of 2, so that

$$x_p[n] = \begin{cases} x[n], & n = 0, \pm 2, \pm 4 \dots \\ 0, & n = \pm 1, \pm 3 \dots \end{cases}$$

and

$$x_d[n] = x[2n]$$

- If $x[n]$ is as illustrated in Figure (a), sketch the sequences $x_p[n]$ and $x_d[n]$.
- If $X(e^{j\omega})$ is as shown in Figure (b), sketch $X_p(e^{j\omega})$ and $X_d(e^{j\omega})$.



2. Fourier transform of decimated signal

Consider the system shown in the figure below, with input $x[n]$ and the corresponding output $y[n]$. The zero-insertion system inserts two points with zero amplitude between each of the sequence values in $x[n]$. The decimation is defined by

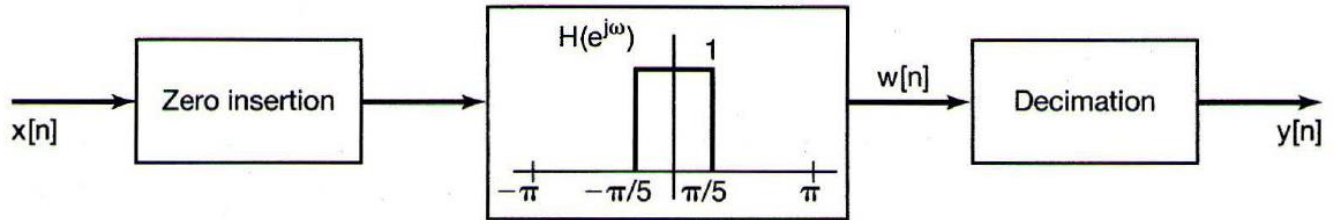
$$y[n] = w[5n]$$

where $w[n]$ is the input sequence for the decimation system. If the input is of the form

$$x[n] = \frac{\sin(w_1 n)}{\pi n}$$

determine the output $y[n]$ for the following values of w_1 :

- $w_1 < \frac{3\pi}{5}$
- $\frac{3\pi}{5} \leq w_1$

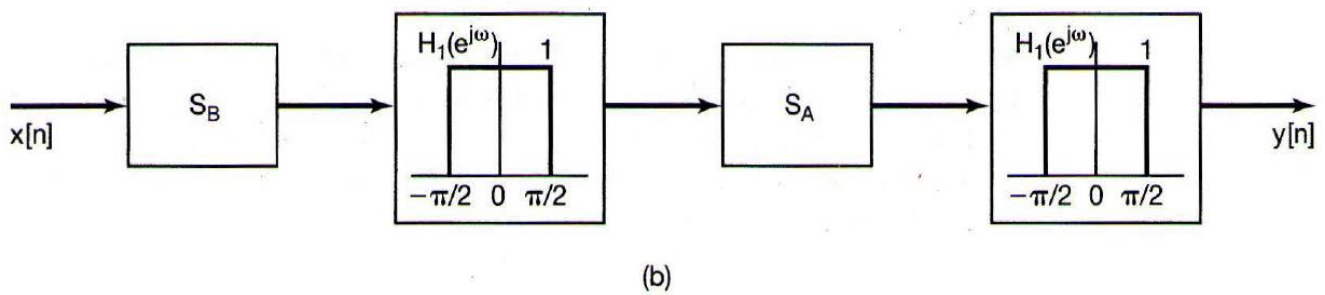
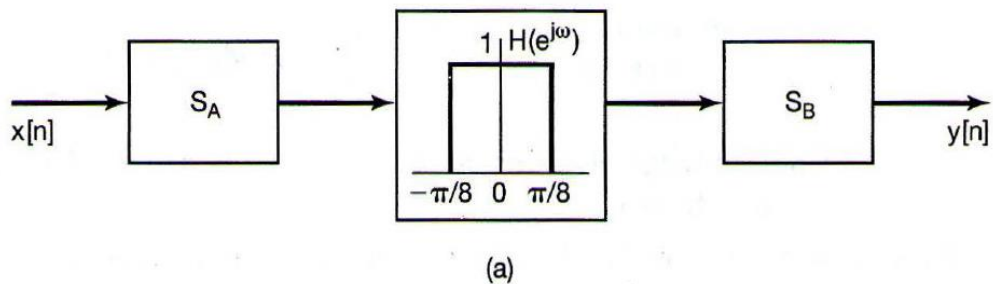


3. Graphical analysis of Decimation and Interpolation function

Two discrete-time systems S_1 and S_2 are proposed for implementing an ideal low-pass filter with cutoff frequency $\pi/4$. System S_1 is depicted in figure (a) below. System S_2 is depicted in the figure (b). In these figures, S_A corresponds to a zero insertion system that inserts one zero after every input sample, while S_B corresponds to a decimation system that extracts every second sample of its input.

(a) Does the proposed system S_1 correspond to the desired ideal low-pass filter?

(b) Does the proposed system S_2 correspond to the desired ideal low-pass filter?



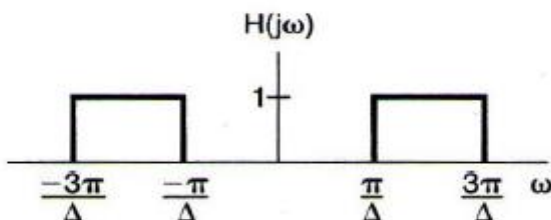
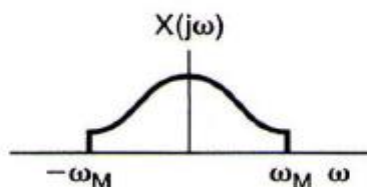
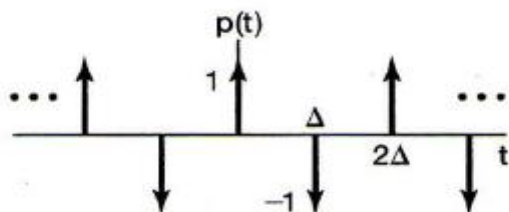
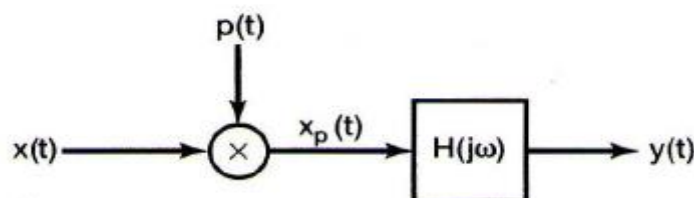
4. Graphical Analysis of Interpolation and Decimation

A real-valued discrete-time signal $x[n]$ has a Fourier transform $X(e^{j\omega})$ that is zero for $3\pi/14 \leq \omega \leq \pi$. The nonzero portion of the Fourier transform of one period of $X(e^{j\omega})$ can be made to occupy the region $|\omega| < \pi$ by first performing up-sampling by a factor of L and then performing down-sampling by a factor of M . Specify the values of L and M .

5. Variation in Sampling Function

Shown in the figure below is a system in which the sampling signal is an impulse train with alternating sign. The Fourier transform of the input signal is as indicated in the figure.

- For $\Delta < \pi/(2\omega_M)$, sketch the Fourier transform of $x_p(t)$ and $y(t)$.
- For $\Delta < \pi/(2\omega_M)$, determine a system that will recover $x(t)$ from $x_p(t)$.
- For $\Delta < \pi/(2\omega_M)$, determine a system that will recover $x(t)$ from $y(t)$.
- What is the maximum value of Δ in relation to ω_m for which $x(t)$ can be recovered from either $x_p(t)$ or $y(t)$?



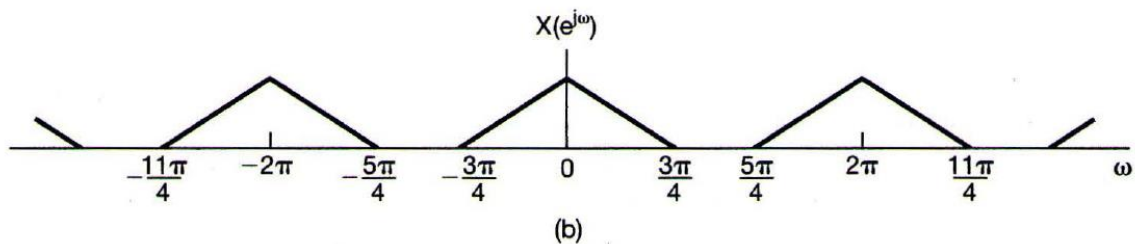
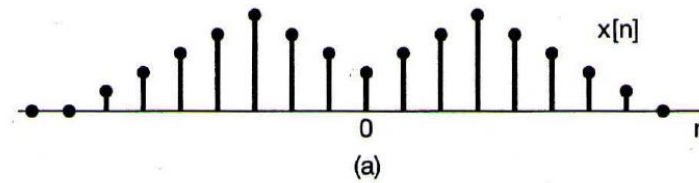
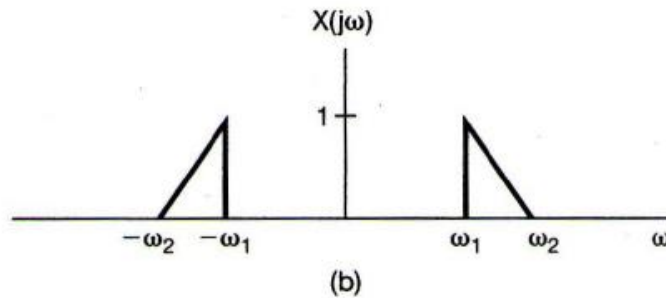
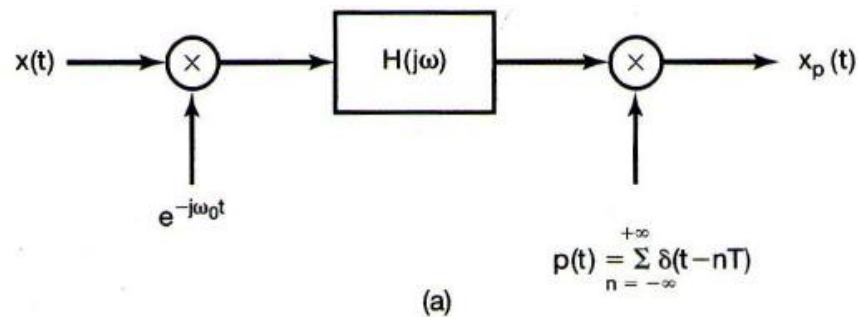
6. Band Pass Sampling

A procedure for band pass sampling and reconstruction, used when $x(t)$ is real, consists of multiplying $x(t)$ by a complex-exponential and then sampling the product. The sampling system is shown in figure(a) below. With $x(t)$ real and with $X(j\omega)$ nonzero only for $\omega_1 < |\omega| < \omega_2$, the frequency is chosen to be $\omega_0 = (1/2)(\omega_1 + \omega_2)$, and the lowpass filter $H_1(j\omega)$ has cutoff frequency $(1/2)(\omega_2 - \omega_1)$.

(a) For $X(j\omega)$ as shown in figure (b), sketch $X_p(j\omega)$.

(b) Determine the maximum sampling period T such that $x(t)$ is recoverable from $X_p(t)$.

(c) Determine a system to recover $x(t)$ from $x_p(t)$.



7. Properties of Sampled signal

Let. $x_c(t)$ be a continuous-time signal whose Fourier transform has the property that

$$x_c(j\omega) = 0 \text{ for } |\omega| \geq 2000\pi$$

A discrete-time signal is obtained. For each of the following constraints on the Fourier transform of $x_d[n]$, determine the corresponding constraint on $X_c(j\omega)$:

- (a) $X_d(e^{j\omega})$ is real.
- (b) The maximum value of $X_d(e^{j\omega})$ over all ω is 1.
- (c) $X_d(e^{j\omega}) = 0$ for $\frac{3\pi}{4} \leq |\omega| \leq \pi$
- (d) $X_d(e^{j\omega}) = X_d(e^{j(\omega-\pi)})$.

Z transforms

8. Z-transform of stable system

We are given the following five facts about a discrete-time signal $x[n]$ with z transform $X(z)$:

- 1. $x[n]$ is real and right-sided.
- 2. $X(z)$ has exactly two poles.
- 3. $X(z)$ has two zeros at the origin.
- 4. $X(z)$ has a pole at $z = \frac{1}{2}e^{j\pi/3}$.
- 5. $X(1) = 8/3$

Determine $X(z)$ and specify its region of convergence.

9. Elementary transformation in z-transform

Let $x[n]$ be a discrete-time signal with z-transform $X(z)$. For each of the following signals, determine the z-transform in terms of $X(z)$:

- a) $\Delta x[n]$, where Δ is the first difference operator defined by

$$\Delta x[n] = x[n] - x[n-1]$$

b) $x_1[n] = \begin{cases} x\left[\frac{n}{2}\right], & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$

c) $x_2[n] = x[2n]$

10. Z-transforms and stability criterion

Determine which of the following z-transforms could be the transfer function of a discrete-time linear system that is not necessarily stable, but for which the unit sample response is zero for $n < 0$. State your reasons clearly.

a) $\frac{(1-z^{-1})^2}{1-\frac{1}{2}z^{-1}}$

b) $\frac{(z-1)^2}{z-\frac{1}{2}}$

c) $\frac{(z-\frac{1}{4})^5}{(z-\frac{1}{2})^6}$

d) $\frac{(z-\frac{1}{4})^6}{(z-\frac{1}{2})^5}$

11. Solution of Difference Equation from z-transform

A sequence $x[n]$ is the output of an LTI system whose input is $s[n]$. The system is described by the difference equation

$$x[n] = s[n] - e^{8\alpha}s[n-8],$$

where $0 < \alpha < 1$.

(a) Find the system function

$$H_1(z) = \frac{X(z)}{S(z)}$$

, and plot its poles and zeros in the z-plane. Indicate the region of convergence.

(b) We wish to recover $s[n]$ from $x[n]$ with an LTI system. Find the system function

$$H_2(z) = \frac{X(z)}{S(z)}$$

such that $y[n] = s[n]$. Find all possible regions of convergence for $H_2(z)$, and for each, tell whether or not the system is causal or stable.

(c) Find all possible choices for the unit impulse response $h_2[n]$ such that,

$$y[n] = h_2[n] * x[n] = s[n]$$